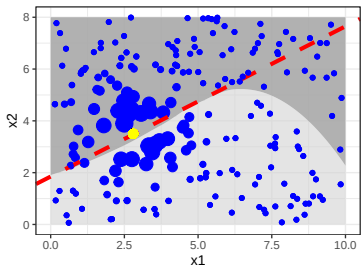
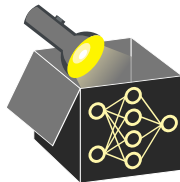


Interpretable Machine Learning

Local Explanations: LIME Pitfalls

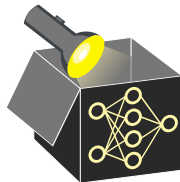


Learning goals

- Learn why LIME should be used with caution
- Possible pitfalls of LIME

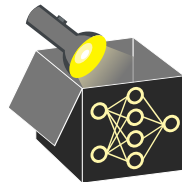
LIME PITFALLS

- LIME is one of the most widely used methods for local interpretability
~> But several papers highlight important (practical) limitations
- Pitfalls arise at multiple levels, which will be discussed in detail:
 - **Sampling** – ignores feature dependencies, risks extrapolation
 - **Locality definition** – kernel width and dist. metrics affect sensitivity
 - **Local vs. global feats** – global signals may overshadow local ones
 - **Faithfulness** – trade-off between sparsity and local accuracy
 - **Hiding biases** – explanations can be manipulated to appear fair
 - **Robustness** – explanations vary for similar points
 - **Superpixels (images)** – instability due to segmentation method



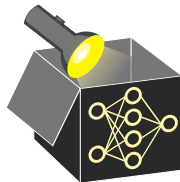
PITFALL: SAMPLING

- **Pitfall:** Common sampling strategies for $\mathbf{z} \in \mathcal{Z}$ ignore feat dependencies
- **Implication:** Surrogate model may be trained on unrealistic points
 $\rightsquigarrow$ Undermines the fidelity and validity of the explanation



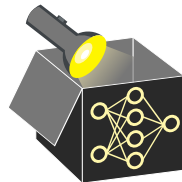
PITFALL: SAMPLING

- **Pitfall:** Common sampling strategies for $\mathbf{z} \in \mathcal{Z}$ ignore feat dependencies
- **Implication:** Surrogate model may be trained on unrealistic points
~> Undermines the fidelity and validity of the explanation
- **Solution I:** Sample locally from the true data manifold $\mathcal{X}$
~> Challenging in high-dimensional or mixed-type data settings
- **Solution II:** Restrict sampling to training data near $\mathbf{x}$
~> Requires enough training data points near $\mathbf{x}$



LIME PITFALL: LOCALITY

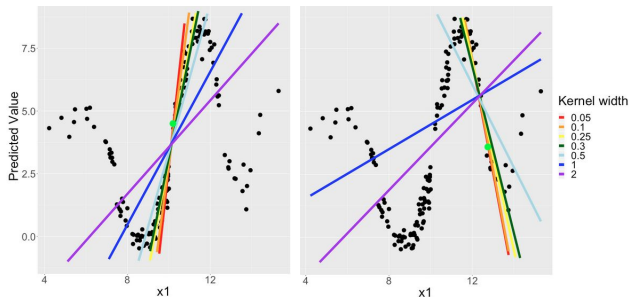
- **Pitfall:** Difficult to define locality (= how samples are weighted locally)
- **Implication:** Local model and explanation quality depend heavily on this weighting, but no principled way exists to choose it
- **Default:** Use exponential kernel as proximity measure between $\mathbf{x}$ and $\mathbf{z}$:
$$\phi_{\mathbf{x}}(\mathbf{z}) = \exp(-d(\mathbf{x}, \mathbf{z})^2 / \sigma^2)$$
 with distance measure d and kernel width σ



LIME PITFALL: LOCALITY

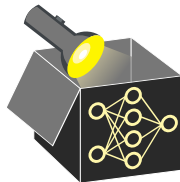
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Example: For 2 obs. (green points), fit local surr. models (lines) using only x_1



Line colors: different kernel widths used for proximity weighting

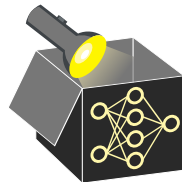
Right: larger kernel widths affect lines more



LIME PITFALL: LOCALITY

► “Kopper et al.” 2019

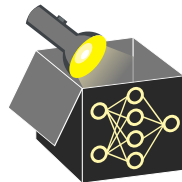
- **Pitfall:** Choice of kernel width (σ) critically influences locality



LIME PITFALL: LOCALITY

► “Kopper et al.” 2019

- **Pitfall:** Choice of kernel width (σ) critically influences locality
- **Implication of edge cases:**
 - *Large* $\sigma \rightarrow$ overemphasize distant points, hurting locality
 - *Small* $\sigma \rightarrow$ too few points may lead to unstable or noisy explanations

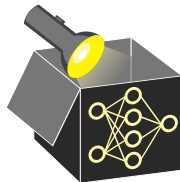


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- **Solution I:** Use Gower similarity directly as weights: $\pi(\mathbf{z}) = 1 - d_{\text{Gower}}(\mathbf{x}, \mathbf{z})$
 - ↪ No kernel width required, but far points still receive (too high) weight
 - ↪ Explanation may reflect more global than local structure
 - ↪ Used in practical LIME implementations

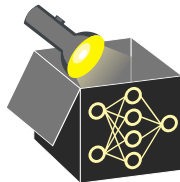
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LIME PITFALL: LOCALITY

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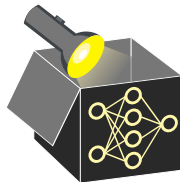
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- **Solution II:** s-LIME adaptively selects σ to balance fidelity and stability ► “Gaudel et al.” 2022



PITFALL: LOCAL VS. GLOBAL FEATURES

► “Laugel et al.” 2018

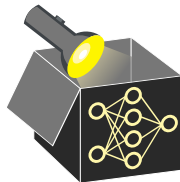
- **Pitfall:** Sampling from entire input space may hide influence of locally relevant feat in favor of globally relevant ones, even for narrow kernels.
- **Feature types:**
 - *Global features* influence predictions broadly across whole input space $\mathcal{X}$
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PITFALL: LOCAL VS. GLOBAL FEATURES

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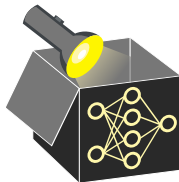
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PITFALL: LOCAL VS. GLOBAL FEATURES

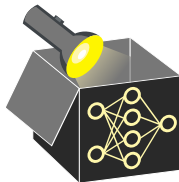
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- **Example:** Decision trees
 - Features near the root impact many instances $\rightarrow$ global
 - Features in lower nodes act locally



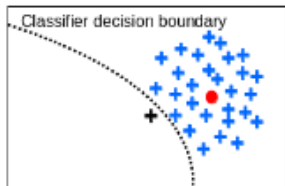
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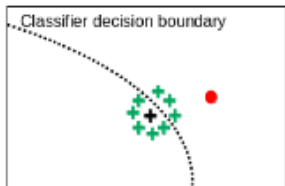


- **Problem:** Sampling around observation to be explained $\mathbf{x}$ may miss decision boundary
- **Solution (LS: Local Surrogate Method):**
 - ❶ Find closest point to $\mathbf{x}$ (red dot) from opposite class (black cross)
 - ❷ Sample around that point to better capture boundary
 - ❸ Train local surrogate using those samples

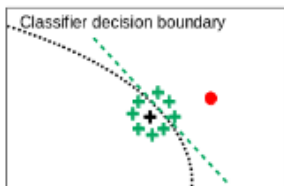
↪ better approximates the local direction of the decision boundary



Step 1: Closest border detection



Step 2: Local sampling

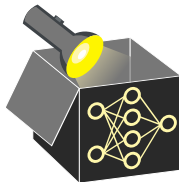


Step 3: Model training

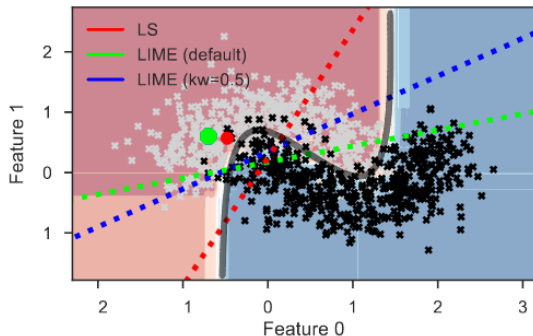
Example: $\mathbf{x}$ (red point), closest point from other class (black cross)

- LIME: What does the model do around this point?
- LS: How does the model change when crossing boundary near this point?

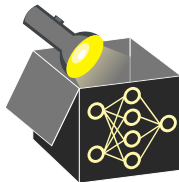
PITFALL: LOCAL VS. GLOBAL FEATS – EXAMPLE



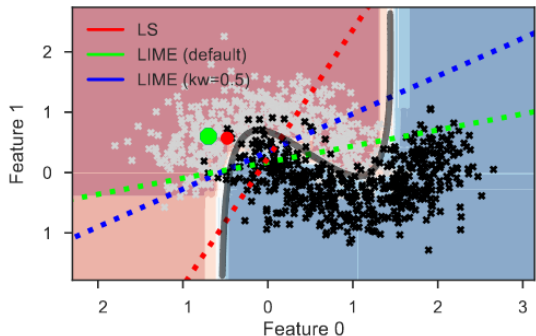
- Random forest (RF) classification on half-moons dataset
- **Background color:** Classification of RF (prediction surface)
- **Black/grey crosses:** training data
- **Green dot:** Obs. to be explained; **Red dot:** nearest opposite-class point
- **Grey curve:** RF's decision boundary; **Dotted lines:** LIME dec. bound.
- **Red line:** Local surrogate (LS) method ► "Laugel et al." 2018



PITFALL: LOCAL VS. GLOBAL FEATS – EXAMPLE



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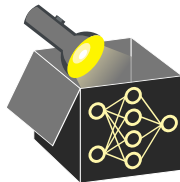
Feature 0 is global;
class always flips when moving left (red) to right (blue)

Feature 1 is local;
class flips only near boundary when moving up/down

Observation: LIME decision boundaries (blue/green) fail to match the steep local bound captured by LS (red)

PITFALL: FAITHFULNESS

- **Problem:** Trade-off between local fidelity vs. sparsity
- **Observation:**
 - Too simple model $\rightsquigarrow$ low fidelity $\rightsquigarrow$ unreliable explanations
 - Complex model $\rightsquigarrow$ high fidelity $\rightsquigarrow$ difficult to interpret surrogate



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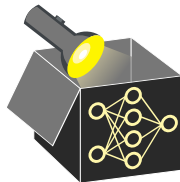
- **Example: Credit data**

- Random forest prediction for $\mathbf{x}$: $\hat{f}(\mathbf{x}) = \hat{\mathbb{P}}(y = \text{bad} \mid \mathbf{x}) = 0.143$
- Sparse LM with 3 features (age, checking.account, duration):

$$\hat{g}_{lm}(\mathbf{x}) = \hat{\theta}_0 + \hat{\theta}_1 x_{age} + \hat{\theta}_2 x_{checking.account} + \hat{\theta}_3 x_{duration} = 0.283$$

- Generalized additive model (with all 9 features) is more complex:

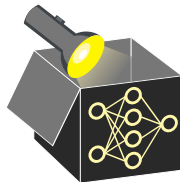
$$\hat{g}_{gam}(\mathbf{x}) = \hat{\theta}_0 + f_1(x_{age}) + f_2(x_{checking.account}) + f_3(x_{duration}) + \dots = 0.148$$



PITFALL: HIDING BIASES

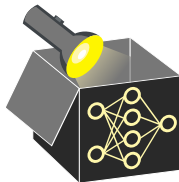
► “Slack et al.” 2020

- **Problem:** LIME samples out-of-distribution (OOD) points, making it exploitable
- **Risk:** Developers can adversarially hide bias in the original model

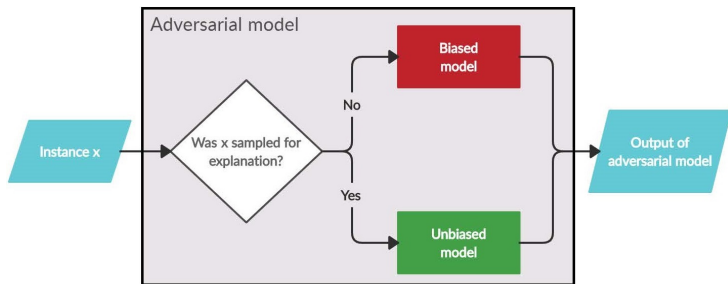


PITFALL: HIDING BIASES

► "Slack et al." 2020



- **Problem:** LIME samples out-of-distribution (OOD) points, making it exploitable
 - **Risk:** Developers can adversarially hide bias in the original model
 - **Attack** with adversarial model:
 - 1 Train a detector to distinguish in-distribution vs. OOD points
 - 2 Use **biased model** for in-distribution inputs (i.e., true predictions)
 - 3 Use **unbiased model** for OOD samples to get LIME explanations
- ~> LIME explanations rely on unbiased model
⇒ hides bias in original model



► [Click for source](#)

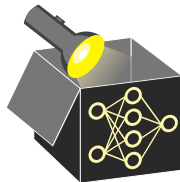
PITFALL: HIDING BIASES

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Key insight: LIME can be fooled if explanations rely on model behavior outside the true data manifold.

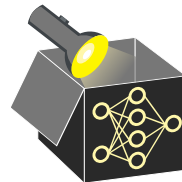
Example: Credit approval

- Biased model uses feats correlated with gender (parental leave duration)
 $\rightsquigarrow$ used to make biased/unfair predictions
- Unbiased model uses only features unrelated to gender for fairness
 $\rightsquigarrow$ used to produce explanations based on unbiased predictions in order to hide bias
- LIME's extrapolated samples trigger the unbiased model
 $\Rightarrow$ explanation appears fair, but original predictions are biased

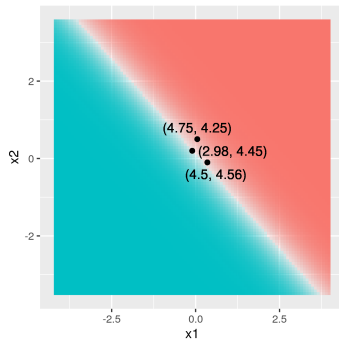


PITFALL: ROBUSTNESS

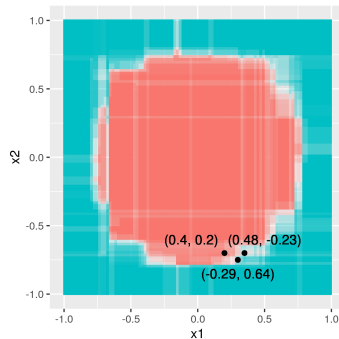
► "Alvarez-Melis, D., & Jaakkola, T." 2018



- **Problem:** Instability of LIME explanations
- **Observation:** Explanations of two very close points could vary greatly
 $\rightsquigarrow$ Variability driven by the stochastic sampling of $\mathbf{z}$ for each explanation
- **Example:**



Linear task (logistic regression).
LIME returns similar coefficients for similar points.

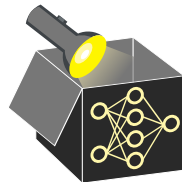


Nonlinear task (random forest).
LIME returns different coefficients for similar points.

PITFALL: DEFINITION OF SUPERPIXELS

► “Achanta et al.” 2012

- **Problem:** LIME relies on superpixels, but their definition differ for image data
- **Observation:** Definition of superpixel influences their size, shape, and alignment



PITFALL: DEFINITION OF SUPERPIXELS

► "Achanta et al." 2012

- **Problem:** LIME relies on superpixels, but their definition differ for image data
- **Observation:** Definition of superpixel influences their size, shape, and alignment
- **Implication:** Specification of superpixel has a large influence on LIME explanations
- **Attack:** Change superpixels as part of an adversarial attack $\rightsquigarrow$ changed explanation

